ECON 662

Energy Economics

University of Michigan

Dynamics: Methodology

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Syllabus

Week 10	10-Mar	Dynamics: Methodology	Rust (1987); Scott (2014)
	12-Mar	Exogeneous Volatility: oil prices, electricity prices…	Kellogg (2014)
Week 11	17-Mar	Structural Regime Shifts	Igami (2018) *Gonzales, Ito & Reguant (2023)*
	19-Mar	Structural Regime Shifts	Boomhower (2019) *(Reduced-form)*
Week 12	24-Mar	Structural Regime Shifts	Fowlie, Reguant, & Ryan (2016)
	26-Mar	Flexible: Concepts review	MWG Ch13&14
Week 13	31-Mar	Information and Dynamics: Learning	Hodgson (2024)
	2-Apr	Information and Dynamics: Lemons	Wang & Xing (2025)
Week 14	7-Apr	Regulation and Incentives: Principal-agent dynamics	Lim, C. S., & Yurukoglu, A. (2018)
	9-Apr	Regulation and Incentives: Principal-agent dynamics	Blundell, Gowrisankaran, & Langer (2020)
Week 15	14-Apr	Summary + Industrial policy	--
	16-Apr	Student Presentation	

Before: organized by industry
Now: organized by **market failures or strategic forces.**
We will briefly discuss how the insights generalize to other industries (or why they may not) at the end of each chapter

Syllabus

- **Today: Methodology**

1. Exogeneous Volatility
   - How do firms respond to oil prices fluctuations, electricity price fluctuations, …

2. Structural shifts and industry evolution
   - How does demand and cost shocks (e.g., environmental regulation) push industry structure evolutions?
   - 3 papers + a summary (review dynamics + introduce principal-agent problems)

3. Information and dynamics

4. Regulation and incentives: Principal-Agent dynamics

Market failures like environmental externality can be corrected by a Pigouvian tax.

Two important other market failures

- **Asymmetric information and hidden type → adverse selection**
- **Unobserved effort and hidden action → moral hazard**

Syllabus

- Today: Methodology

1. Exogeneous Volatility
   - How do firms respond to oil prices fluctuations, electricity price fluctuations, …
2. Structural shifts and industry evolution
   - How does demand and cost shocks (e.g., environmental regulation) push industry structure evolutions?
   - 3 papers + a summary (review dynamics + introduce principal-agent problems)
3. Information and dynamics
   - Firm learning and information externality. What's the optimal design?
   - Consumer learning and quality reputation externality. What's the optimal design?
4. Regulation and incentives: Principal-Agent dynamics
   - Set up the regulator's problem and estimate the game between regulators and polluting firms. What if firms manipulate inspection frequency? What's the optimal auditing intensity?

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Industrial policy is a broad term, but many of its underlying mechanisms are topics we will discuss throughout the course.
I will give a brief summary in the last Lecture 25.

Goals: for you to write the version 0 of any dynamic paper

1. Be clear about the key economics and forces you want to capture. Choose the right amount of structure and abstract from other factors.

   - "This reduced form model **has just enough structure to capture** the differences in responsiveness to capital costs and variable costs across units. " – Fowlie (2010) AER

   - "This simple model is **parsimonious yet allows us to examine the core of our research question**: the equilibrium solar investment with and without market integration." – Gonzales, Ito & Reguant (2023) EMCA

2. Understand the simplest methods, and the intuition behind more advanced approaches — enough to ***write Version 0 of any paper***.

Why are dynamics important?

- Questions we can answer:
  - How do firms make long-term decisions such as entry, exit, investment, quality upgrading, …
  - How does competition shape market structure evolution?
  - To what extent is industry growth path-dependent?
  - Optimal policy to push the industry to some good equilibrium?

- Examples:
  - Industry evolutions: Electricity, Oil, EV, game between firms
  - Regulation and dynamics: the game between regulator and firms

Roadmap

- **Workhorse Model**
  - Math set up and properties
  - Algorithm to solve to model: *with all known parameter values, how do we get agents' (consumers/firms) optimal decisions?*

- **Estimation Methods (4 common ones)**

  With observed data and assuming the data is generated from sophisticated agent decisions, how to backout the primitive parameter values?

- **Link the model with economic examples**

Model

1. Model setup, definitions, …
2. Math tricks

Model Setup

- Bus engine replacement is an example of the "optimal stopping" problem. Trade off:
  - There are large fixed costs associated with stopping (replacement cost)
  - Future maintenance costs keep growing as mileage accumulate if not replaced
  - When to replace?

- Replace when mileage is larger than a threshold

Model Setup

- Discrete time t ∈ 0, 1, ...
- State variable: the bus engine's mileage Xt and replacement cost RCt

$$u(x_t, RC_t, a_t, \epsilon_t|\theta_1) = \begin{cases} c(x_t|\theta_1) + \epsilon_t^0 & \text{if } a_t = 0 \\ -RC + \epsilon_t^1 & \text{if } a_t = 1 \end{cases}$$

  - Cost shocks ~ i.i.d. Type 1 extreme distribution
- Optimal choice: replace if and only if get some certain states a(x_t, RC_t, ε_t)
  - If the random draw is high enough
  - If mileage is too high to pay the maintenance cost
  - If price today is low enough

Rust 1987 Model Analysis

- ***State variables:*** $S = (x_t, RC_t)$ *mileage, replacement cost*
- ***Actions a***: replace ∈ {0,1}
- ***Per-period Payoff (S, a, shocks)***
  - ***Random shocks:*** cost shocks ε_i

$$u(x_t, RC_t, a_t, \epsilon_t|\theta_1) = \begin{cases} c(x_t|\theta_1) + \epsilon_t^0 & \text{if } a_t = 0 \\ -RC + \epsilon_t^1 & \text{if } a_t = 1 \end{cases}$$

- ***State Transition:***
  - ***Exogeneous***: $RC_{t+1} \sim f(RC_{t+1}|RC_t, \theta_2^{RC})$;
  - ***Endogenous***: $x_{t+1} \sim f(x_{t+1}|a_t,\ x_t, \theta_2^x)$

Replace:

Cost	RC_t
Benefit	$C(x_{t+1}, \theta_1)$in the future

- Key dynamic force:

  How action respond to expected future prices and mileages

- Key primitive:

  θ_1 determines cost saving, θ_2 determines by how much the action change mileage

Equilibrium Concept and outcome

- Markov Perfect Equilibrium
  - Markovian replacement policy: action only depends on the current state, no historical info.
  - **Policy function $a(x_t, RC_t, \varepsilon_t)$ and value function $V(x_t, RC_t, \varepsilon_t)$**

$$V(x_t, RC_t, \epsilon_t) =$$

$$\max \begin{cases} c(x_t|\theta_1) + \epsilon_t^0 + \beta \cdot \overbrace{\int_{(x,rc,\epsilon)} V(x_{t+1}, RC_{t+1}, \epsilon_{t+1}) \cdot \boxed{f(S_{t+1}|S_t, a=0)ds}}^{EV_{a=0}} & \text{if } a_t = 0 \\ -RC_t + \epsilon_t^1 + \beta \cdot \underbrace{\int_{(x,rc,\epsilon)} V(x_{t+1}, RC_{t+1}, \epsilon_{t+1}) \cdot f(S_{t+1}|S_t, a=1)ds}_{EV_{a=1}} & \text{if } a_t = 1 \end{cases}$$

Equilibrium Concept and outcome

- Markov Perfect Equilibrium
  - Markovian replacement policy: action only depends on the current state, no historical info.
  - **Policy function $a(x_t, RC_t, \varepsilon_t)$ and value function $V(x_t, RC_t, \varepsilon_t)$**
- For simplicity:

$$\delta_j(s) = u_j(s) + \beta EV_j(s)$$

$$EV_j(s) = TP(j,s)_{1-by-|S|} \times V_{|S|-by-1} = \sum_{s' \in S} TP(s'|j,s) \cdot V(s')$$

  - Probability of replacement in state s P(s)= … ?
  - Value of state V(s) = … ?

$$V = ln(\sum_j exp(\delta_j)) + \gamma,$$

Note:
1 understanding the matrix format is important to coding and to your problem set
2 clarify notation of V

Solve the model $V(s)$ and $P(s)$

Get optimal policy and value function, given parameters

1. Calculate transition probabilities

2. Calculate $V(s)_{|S|-by-1}$ and $P(s)_{|S|-by-1}$
   - Guess a vector value of $V(s)_{|S|-by-1}$
   - Get $EV_j(s) = TP(s,j) \times V(s)$ and δ_j
   - Update $V = ln(\sum_j exp(\delta_j)) + \gamma,$ A Fixed Point (FXP) algorithm

3. After $V(s)$ converges, calculate $P(s)$

Properties: *you need to know the intuition but not formal proof*
- Existence
- Uniqueness, yes in this model
- Monotonicity *(not that important)*
  If u(x) is monotonic, V(x) is also monotonic
  Linearity? Usually not preserved

Assumptions

- Markovian (we'll discuss more about this assumption in week 14)
- Conditional independence:
  - ϵ_t is i.i.d., no unobserved serial correlation
  - ϵ_{t+1} is not correlated with x_{t+1}
  - Transition: S_{t+1} does not depends on ϵ_t
  - So we don't have to include ϵ_t in the state variable

Standard in all models

- ***State Transition:***
  - ***Exogeneous***: $RC_{t+1} \sim f(RC_{t+1}|RC_t, \theta_2^{RC})$;
  - ***Endogenous***: $x_{t+1} \sim f(x_{t+1}|a_t,\ x_t, \theta_2^{x})$

- All state variables are observed, all unobserved factors affecting action is temporary

CONDITIONAL INDEPENDENCE ASSUMPTION (CI): *The transition density of the controlled process* $\{x_t, \varepsilon_t\}$ *factors as*

$$p(x_{t+1}, \varepsilon_{t+1}|x_t, \varepsilon_t, i, \theta_2, \theta_3) = q(\varepsilon_{t+1}|x_{t+1}, \theta_2)p(x_{t+1}|x_t, i, \theta_3). \tag{4.7}$$

Assumptions

- Without ϵ, intuitively, what would be the optimal maintenance policy?
  - Maintain when … ?
  - Does all buses follow this rule?
  - How important is this threshold or math sense in empirical model?

We can identify the scale of ϵ_{it} because … ?
But in BLP … ?

- ϵ: to match and explain the data better (as in all empirical models)
  - Some paper interpret it as "unobserved heterogeneity", agent's "private information", and even denote it with x_i, θ_i, … Usually this is more about the econometrics
  - This is assumed to be private information, so other players don't know what econometricians do not know.
  - *Note 1: if other players know, we need a different estimation method (complete information entry game, …)*
  - *Note 2: we need some modification if the "unobserved heterogeneity" is persistent, usually it's not*

Thursday's paper: x_{it} in Kellogg 2014

Three common distributional assumptions

- Type I extreme distribution (only in discrete choice)
  - We'll start here

- Normal distribution (both discrete and continuous choices)
  - There will be one or more papers, I will explain more there.

- Exponential distribution (both discrete and continuous choices)
  - We will not cover any paper, but Luming's JMP uses this distributional assumption

- **Reasons:**
  - All have closed-form CDFs → Choice probabilities
  - Which one to use: almost purely math/coding reasons

Estimation

With observed data and assuming the data is generated from a dynamic model, how to estimate the primitive parameter values?

Note: Today we'll focus on the algorithm and implementation, we'll talk about identification when discussing each paper

Approach 1: Rust 1987 Nested FXP

- Step 1: Estimate Transition Matrix
  - Both exogeneous and endogenous (conditional on action and state)

- Step 2: Estimate V and P

Guess θ

- Guess a vector value of $V(s)_{|S|-by-1}$
- Get $EV_j(s) = TP(s,j) \times V(s)$ and δ_j
- Update $V = ln(\sum_j exp(\delta_j)) + \gamma,$

A Fixed Point (FXP) algorithm

3. After $V(s)$ converges, calculate $P(s)$

- Step 3: Calculate likelihood given current θ

Hotz Miller 1993

Approach 2: Use more information from data

- Can we get rid of the nested fixed point?
  - Given a guess of parameter (inner loop), why do we need a fixed point algo to get V?
  - V shows up in both the left and the right hand side of $V = ln(\sum_j exp(\delta_j)) + \gamma,$
  - Other expressions for calculating V?

$$V = \sum_{j \in J} Pr_j^* \cdot (\delta_j + \underbrace{E[\varepsilon_j | \text{ option } j \text{ is selected by agent } i]}_{\text{denoted } \phi_j})$$

$$\phi_j = \gamma - lnPr_j^*,$$

- P is observed!
  - V … how about we move them to the same side of the equation?

Hotz Miller 1993

Approach 2: Use more information from data

$$\underbrace{V(s)}_{|S|\text{-by-1 vector}} = \left[I_{|S|-by-|S|} - \beta \sum_j P_j(s) \cdot TP(j,s) \right]^{-1} \times \left[\sum_j P_j(s) \cdot (u_j(s) + \phi_j(s)) \right]$$

The HM Inversion

- What is observed and what is unobserved in the data?
- Given any guess of parameters, how can we get V(s)?

Hotz Miller 1993

Approach 2: Use more information from data

- Step 1: Estimate Transition Matrix
  - Both exogeneous and endogenous (conditional on action and state)
- Step 2: Estimate conditional choice probability from data (CCP)
  - At each mileage, each RC value, what's the probability of the buses replace its engines
- Step 3: Guess θ, approximate the corresponding value function
  - V $\leftarrow \widehat{CCP}, \theta$, using HM inversion
  - Get P from the approximated V
- Step 4: Calculate likelihood given current θ

Approach 3: Forward simulation

- Step 1: Estimate Transition Matrix
  - Both exogeneous and endogenous (conditional on action and state)
- Step 2: Estimate conditional choice probability from data (CCP)
  - At each mileage, each RC values, what's the probability of the buses replace its engines
- Step 3: Guess θ, approximate the corresponding value function
  - At any state s:
    - Random draw the action based on CCP
    - Random draw next stage's state variable based on TP and ϵ
    - Get the utility from t+1, t+2, ….
    - Calculate V(s) by adding up all future utilities
  - Get P from the approximated V
- Step 4: Calculate likelihood given current θ

Note:
1. This usually is not applied to SAD
2. This captures the spirit of a more advanced method in BBL and POB, but I want to introduce this idea here.
3. This only explains 50% of the beauty and smartness of BBL and POB

Scott 2014

Approach 4: Use even more information from the data

- If we observe both x and V, we can just run a simply regression

$$lnPr_j - lnPr_k = \underbrace{\theta_1 \cdot (x_j - x_k)}_{\text{different per-period payoff e.g., charge your EV or not}} + \underbrace{\beta(EV_j - EV_k)}_{\text{You go to different states in the next period}}$$

  - Can we construct future value difference with some observables?
  - Can observed choice probabilities in state x imply the value of state x, V(x)?
  - Hint:

$$V = \delta_j - lnPr(\delta_j) + \gamma \qquad \forall j \in \mathscr{J},$$

Scott 2014

Approach 4: Use even more information from the data

- If we observe both x and V, we can just run a simply regression

$$lnPr_j - lnPr_k = \underbrace{\theta_1 \cdot (x_j - x_k)}_{\text{different per-period payoff e.g., charge your EV or not}} + \underbrace{\beta(EV_j - EV_k)}_{\text{You go to different states in the next period}}$$

- Yes, use more math tricks to construct the difference in EV with observed P
  - Intuition: Pr(replace) at battery level [10,20] - Pr(charge) at [70-80] should reflect V[10]-V[70]
  - Math: $V(x) = -lnPr_{charge}(x) + \delta_{charge}(x) + \gamma\,, \quad \forall j, \forall x$ *(from the cheat sheet)*
    - Note: $\delta_{charge}(x) = u(x, charge) + \beta\, V(full\ battery), \quad \forall x.$
    - $\delta_{charge}(10) - \delta_{charge}(70) = u(10, charge) - u(70, charge)$, no unobservable!
  - **Termination Action Assumption**

Scott 2014

Approach 4: Use even more information from the data

- If we observe both x and V, we can just run a simply regression

$$lnPr_j - lnPr_k = \underbrace{\theta_1 \cdot (x_j - x_k)}_{\text{different per-period payoff e.g., charge your EV or not}} + \underbrace{\beta(EV_j - EV_k)}_{\text{You go to different states in the next period}}$$

- Yes, use more math tricks to construct the difference in EV with observed P

$$EV_j - EV_k = V^{rlz}_{t+1}(a_t = j) - V^{rlz}_{t+1}(a_t = k) + e^{rlz}_j - e^{rlz}_k$$

$$= [-lnPr_j(s1) + \delta_j(s1) + \gamma] - [-lnPr_j(s2) + \delta_j(s2) + \gamma] + \Delta e^{rlz} \qquad \forall j \in \mathcal{J}$$

$$= \ldots \textit{any action gives exact the same EV no matter which state I am in?} \ldots$$

$$= -lnPr_j(s1) + lnPr_j(s2) + u(s1) - u(s2) + \Delta e^{rlz}$$

Scott 2014

Approach 4: Use even more information from the data

- If we observe both x and V, we can just run a simply regression

$$lnPr_j - lnPr_k = \underbrace{\theta_1 \cdot (x_j - x_k)}_{\text{different per-period payoff e.g., charge your EV or not}} + \underbrace{\beta(EV_j - EV_k)}_{\text{You go to different states in the next period}}$$

- Yes, use more math tricks to construct the difference in EV with observed P

$$\frac{lnPr(a=0,s)}{lnPr(a=1,s)} + \beta \underbrace{\frac{lnPr(a=1|s'_{a_t=0})}{lnPr(a=1|s'_{a_t=1})}}_{\text{Approximate future value difference}} = c(x|\theta_1) - RC + \Delta e^{rlz}_{t+1}$$

  - If $c(x|\theta_1) = -\theta_1 \cdot x$, …

The Generic Approach (for 2,3,4)

- Step 1: Estimate Transition Matrix
  - Both exogeneous and endogenous (conditional on action and state)

- Step 2: Estimate conditional choice probability from data (CCP)
  - Need a modification if choice is continuous (e.g., investment), we will see it next week

- Step 3: Approximate the corresponding value function
  - V ← $\widehat{CCP}, \theta$, using HM inversion or forward simulation

- Step 4: Calculate likelihood or match moments given current θ

Approach 4 Scott 2014 constructs an OLS equation

Economic examples

What can we analyze with a dynamic model?

- Cool simulations that replicate industry growth path!
- But what are the economics?
  - Type I: Agent's respond to exogeneous fluctuations, regulatory changes, … Optimal Stopping
    - Oil extraction, investment, …
    - Firm's choice depends on some state variable (e.g., prices)
  - Type II: Agent's choice affect state evolution (not just non-invertible) Optimal Control
    - Market structure evolutions, …
    - Strategic interaction among agents
  - Most cases: a combination of both

Bailey, M. R., Brown, D. P., Myers, E., Shaffer, B., & Wolak, F. A. (2025). Electric vehicles and the energy transition: Unintended consequences of time-of-use pricing. *American Economic Review: Insights*, 7(4), 550-566.

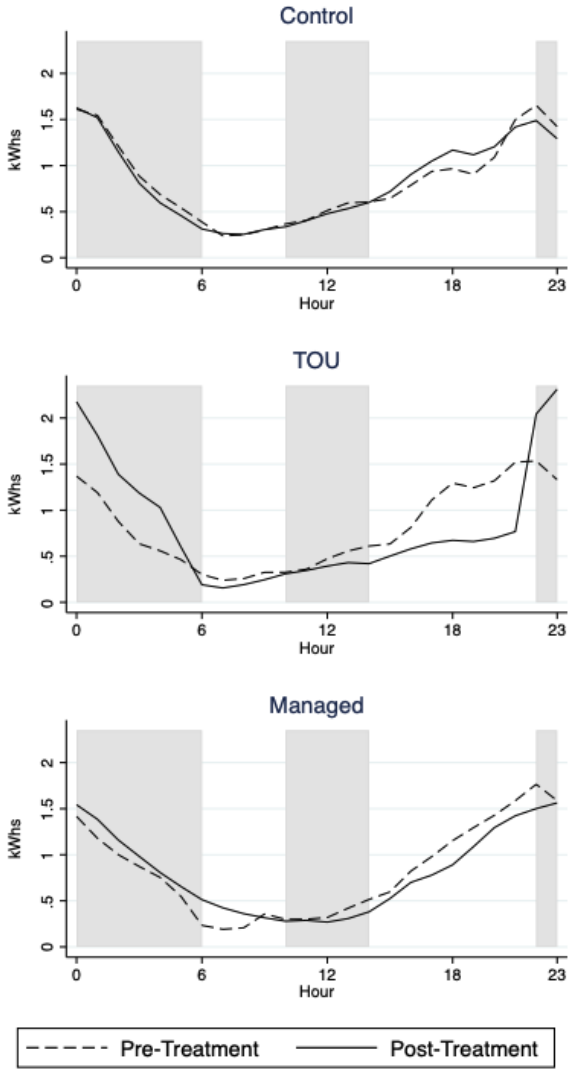
Examples: Exogeneous policies/price fluctuations

- EV Charing Problem
  - Decision: charge now or wait
  - Extensions? Study finds EVs create shadow peak, how to solve the problem?
  - Why dynamics?

- Battery investment:

  Butters, R. A., Dorsey, J., & Gowrisankaran, G. (2025). Soaking up the sun: Battery investment, renewable energy, and market equilibrium. Econometrica, 93(3), 891-927.

- Deforestation problem (Scott 2014 and many follow up papers)
  - Decision: land usage in each year: forest, farm for cows, farm for rice
  - Agricultural products (dairy and rice) prices in each year
  - Why dynamics?



(a) Average Charged kWh

More problems

- Quality upgrading competition
  - Nvidia and AMD product quality (in mm) in each year
  - …
- Regulator and polluter dynamics
  - …

Note: The model and methods we talked about today (except for Approach 3) are for single-agent dynamics

A dynamic model with 2 agents' *(Nvidia and AMD)* strategic interactions are NOT fundamentally different *(week 13, after you've been familiar with single agent dynamics)*

- The solution method follows the same ideas, same 4 steps, either Approach 1 or Approach 3
- Computationally much much more expensive, sometimes multiple equilibria

Understand a dynamic model

- The first thing to do:
  - What are the state variables
  - What is the dynamic action
  - What are exogeneous evolutions?
    - Oil/gas prices, sometimes quality, …
  - What are endogenous evaluations?
    - Market structure, …

Kellogg 2014 Set up

- Risk-neutral, price-taking oil production firm, oil prices follow a known stochastic process
- **Timing**
  - Well productivity realize.
  - Firm decide if to drill.
  - If so, pay the dayrate D_t, and the well produces oil for approximately 20 months
  - If not, wait for future opportunities
  - State transition: oil price and dayrate changes

- Two sources of uncertainty:

  The dynamic stochastic oil price process and the productivity realization

Question for you: does the productivity x_i have any dynamic implications in his setting?

Kellogg 2014 Model Analysis

Please compare his argument for the x_i specifications and today's Page 17

- ***State variables:*** $S = (P_t, D_t, \sigma_t)$ oil price, drilling cost (day rate), and vol
- ***Actions a***: drilling ∈ {0,1}
- ***Per-period Payoff (S, a, shocks)***
  - ***Random shocks:*** productivity x_i

$$\pi_{it} = \pi_i(P_t, D_t) = r_i P_t - c_i - d_i D_t.$$

$$= \overline{C}_i(x_i P_t - \tilde{c} - \tilde{d} D_t).$$

- ***State Transition:***
  - ***Exogeneous***: $P_{t+1} = f(P_t, \sigma_t)$; $D_{t+1} = f(D_t, \sigma_t)$; $\sigma_{t+1} = f(\sigma_t)$;

    Each firm believe: $ln\sigma = \overline{ln\,\sigma} + \beta \cdot ln\,\sigma^d$
  - ***Endogenous***: --

Drilling:

Cost Waiting value (.. Wait for what..?...)

Benefit ?

- Key dynamic force:

  ?
- Key primitive:

  ?

Some parameters are just for matching the reality

Appendix

You might have seen dynamic models, especially theoretical ones …

First-year macro dynamic models

- ***Uniqueness and Existence:*** Contraction Mapping
- ***Preserve monotonicity***: if u(x) is monotonic, V(x) is
- ***Preserve concavity***, special case: linearity
- ***Solution method***: sometimes analytical
- A set of assumptions on the domain (compact, closed, …), the range, the function (C1, C2, bounded, …), …

Empirical IO

- ε, explains the difference between observed action and model predicted optimal action
- More precise and flexible u(x) specification: *Micro economics insights need a precise modeling of frictions*
- Discrete choice *(we will talk about continuous choice)*
- We solve V by simulations, usually no closed form solutions

The statements represent my own views with my limited knowledge about macro. Any errors are my responsibility